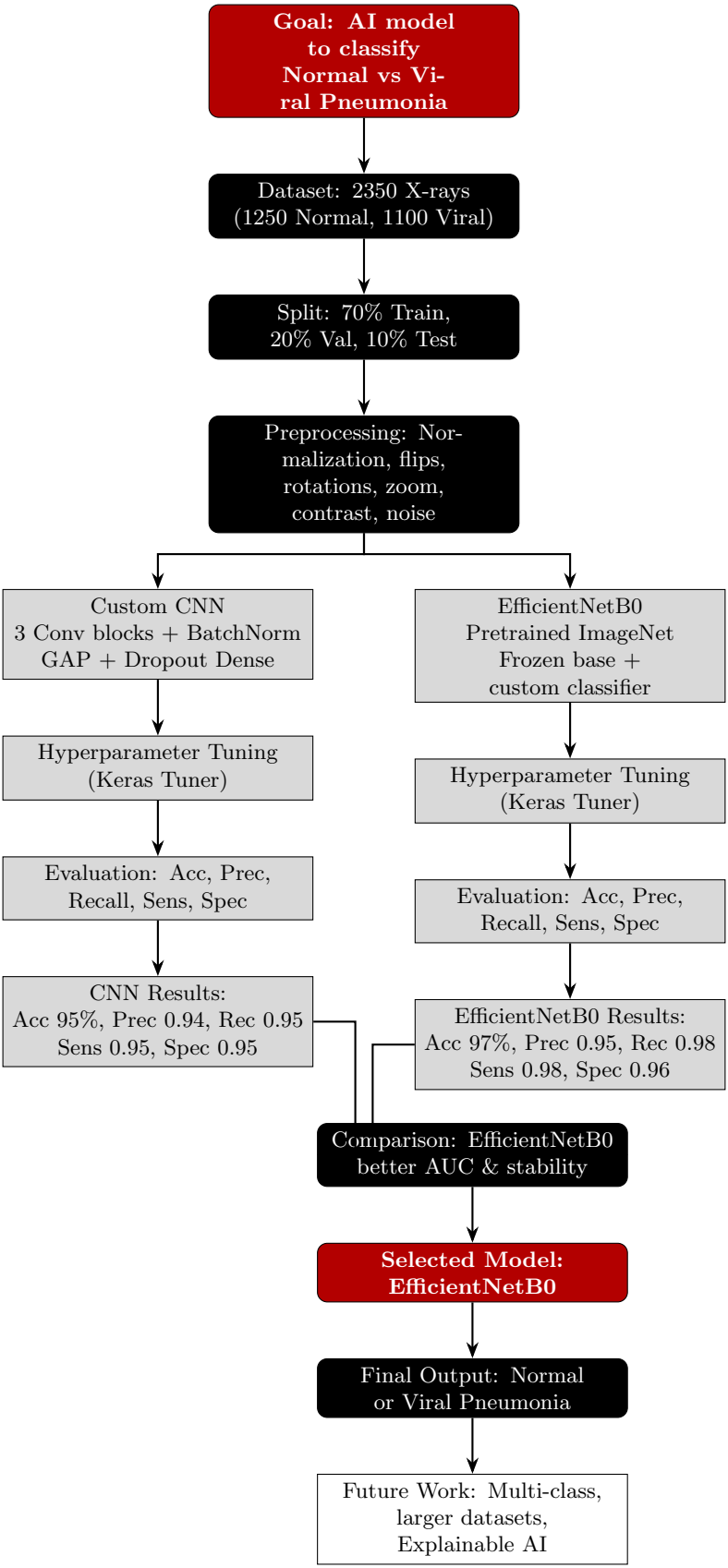
Goal: AI model to classify Normal vs Viral Pneumonia
Dataset: 2350 X-rays (1250 Normal, 1100 Viral)
Split: 70% Train, 20% Val, 10% Test
Preprocessing: Normalization, flips, rotations, zoom, contrast, noise
Custom CNN 3 Conv blocks + BatchNorm GAP + Dropout Dense
EfficientNetB0 Pretrained ImageNet Frozen base + custom classifier
Hyperparameter Tuning (Keras Tuner)
Hyperparameter Tuning (Keras Tuner)
Evaluation: Acc, Prec, Recall, Sens, Spec
Evaluation: Acc, Prec, Recall, Sens, Spec
CNN Results: Acc 95%, Prec 0.94, Rec 0.95 Sens 0.95, Spec 0.95
EfficientNetB0 Results: Acc 97%, Prec 0.95, Rec 0.98 Sens 0.98, Spec 0.96
Comparison: EfficientNetB0 better AUC & stability
Selected Model: EfficientNetB0
Final Output: Normal or Viral Pneumonia
Future Work: Multi-class, larger datasets, Explainable AI